

RESEARCH ARTICLE | DECEMBER 03 2025

Optimizing the dynamic performance of galvanometric scanners through advanced trajectory planning for laser material processing

Special Collection: [Proceedings of the International Congress of Applications of Lasers & Electro-Optics \(ICALEO 2025\)](#)

Pawel Garkusha  ; Felix Harst  ; Michael F. Zaeh 



J. Laser Appl. 38, 012006 (2026)

<https://doi.org/10.2351/7.0001938>



Articles You May Be Interested In

Evaluation of a galvanometric scanner for rapid tuning of CO₂ lasers

Rev. Sci. Instrum. (March 1989)

Dissimilar laser welding of copper and aluminum alloys in multilayer configuration for battery applications

J. Laser Appl. (October 2021)

Comparative analysis of the potential of state-of-the-art lasers and new prototypic high-power beam sources for cutting nonmetals

J. Laser Appl. (April 2020)



Optimizing the dynamic performance of galvanometric scanners through advanced trajectory planning for laser material processing

Cite as: J. Laser Appl. 38, 012006 (2026); doi: 10.2351/7.0001938

Submitted: 13 July 2025 · Accepted: 6 November 2025 ·

Published Online: 3 December 2025



Pawel Garkusha,^{a)} Felix Harst, and Michael F. Zaeh

AFFILIATIONS

TUM School of Engineering and Design, Institute for Machine Tools and Industrial Management (iwb),
Technical University of Munich, Boltzmannstrasse 15, Garching 85748, Germany

Note: Paper published as part of the special topic on Proceedings of the International Congress of Applications of Lasers & Electro-Optics 2025.

^{a)}Author to whom correspondence should be addressed; electronic mail: pawel.garkusha@iwb.tum.de

ABSTRACT

Galvanometric scanners constitute fundamental actuators in precision laser manufacturing, yet inertia-induced tracking errors impose severe throughput limitations for μm -scale positioning tasks. This work presents a software-based trajectory predistortion that compensates for scanner dynamics without hardware modification, enabling immediate deployment across existing industrial systems. Model-based trajectory optimization uses regularized inversion with Butterworth filtering to generate pre-compensated command signals that counteract inherent system lag. Experimental validation demonstrates settling time reductions of 49.2% for $30\text{ }\mu\text{m}$ movements and 17.3% for mm-scale movements. The tracking error decreased by precisely 50% from 200 to $100\text{ }\mu\text{s}$, independent of the movement amplitude. Industrial implementation for titanium porous transport layer fabrication achieved 38.1% throughput enhancement. This software-only approach requires no hardware investment and integrates seamlessly with existing control systems. The methodology generalizes to diverse scanner-based applications, where positioning dynamics constrain manufacturing throughput. Widespread adoption could significantly accelerate production across multiple industries utilizing galvanometric scanning systems, as the technique can maximize the performance from installed equipment bases without hardware replacement or extended qualification procedures.

Key words: galvanometric scanner, trajectory optimization, model inversion, laser material processing, tracking error reduction

© 2025 Author(s). All article content, except where otherwise noted, is licensed under a Creative Commons Attribution (CC BY) license (<https://creativecommons.org/licenses/by/4.0/>). <https://doi.org/10.2351/7.0001938>

I. INTRODUCTION

Galvanometer-based laser scanning systems face fundamental physical constraints imposed by mechanical inertia, thermal drift, and the inherent trade-off between scanning speed and positioning accuracy. These limitations have driven significant innovations in control algorithms and alternative scanning architectures. Model-driven optimization approaches have demonstrated a 50% reduction in settling time through systematic proportional-integral-derivative (PID) parameter tuning alone,¹ while iterative learning control implementations have achieved 73-fold reductions in the root mean square tracking error at scanning rates exceeding 4000 lines per second.² Adaptive inverse control strategies have extended the usable bandwidth by 220%,

pushing the 3 dB damping limits beyond 6 kHz.³ Concurrently, alternative scanning technologies have emerged to circumvent mechanical limitations entirely: micro-electro-mechanical system-based solutions integrate scanning functionality into sub-20 mm packages while operating at 100 kHz with minimal power consumption;⁴ and acousto-optic deflectors enable MHz-rate beam steering, limited only by the acoustic propagation velocity, with recently demonstrated power handling capabilities exceeding 20 GW.⁵ The throughput limitations of single-beam systems have been addressed by developing massively parallel architectures capable of achieving over 1000 simultaneous beams.⁶

While substantial advances in control algorithms, alternative scanning technologies, and parallel processing have increased the

productivity of ultrashort pulse laser material processing, a complementary approach exists that remains largely unexploited—intelligent trajectory planning. The aforementioned innovations primarily address hardware limitations and control system dynamics, yet the fundamental scanning trajectories employed have remained essentially unchanged since the inception of galvanometer scanning. The novelty of this work lies in presenting a transparent and accessible model-based inverse pre-compensation strategy designed specifically for improving the dynamic performance of existing scanner hardware. The proposed method significantly reduces the tracking error, the key metric for dynamic performance, thereby enabling higher processing speed and accuracy without requiring hardware modifications.

This work demonstrates that trajectory predistortion, implemented through regularized model inversion, can yield substantial performance gains from existing galvanometer systems. By pre-compensating for known system dynamics, a 50% reduction in the tracking error is achieved. This software-only approach requires no hardware modifications, offering immediate deployment potential across the installed base of industrial laser scanner systems.

To validate the proposed performance improvements, point-by-point percussion drilling for the fabrication of porous transport layers (PTLs) in proton exchange membrane electrolyzers was selected as a representative application.⁷ This process imposes particularly stringent demands on scanner dynamics, as it involves rapid repositioning between discrete hole locations and requires complete settling within tight positional tolerances before each laser pulse. Any residual oscillation or overshoot directly translates into misaligned or geometrically distorted holes, thereby limiting part quality.⁸ Since the settling time constitutes the principal bottleneck in process speed, reducing dynamic tracking error and overshoot directly shortens this interval, enabling both higher throughput and improved machining precision. It is important to note that this work isolates and addresses only the motion behavior of the galvanometric scanner; effects arising from laser source characteristics, power settling behavior, and scanner communication protocols were explicitly not considered.

II. METHODOLOGY

A. Experimental setup

The experimental system comprised a Raylase Focushifter RD-14 galvanometer scanner integrated with a SPICE-3 control board operating at 100 kHz update rate. Position feedback utilized integrated optical encoders with 20-bit resolution, corresponding to $0.06\mu\text{rad}$ angular resolution over $\pm 0.393\text{ rad}$ mechanical scan angle. Scanner specifications include a maximum angular velocity of 50 rad/s and a nominal tracking error of $200\mu\text{s}$. The experimental data were acquired directly from the scanner's optical encoder feedback system, with position readout latency below $1\mu\text{s}$, negligible compared to the measured tracking errors of $100\text{--}200\mu\text{s}$.

B. Performance metrics

Settling time was defined as the time interval from initiation of a position command until the scanner output enters and remains within a $\pm 1\mu\text{m}$ tolerance band around the target position.

For $30\mu\text{m}$ movements, this tolerance represents approximately 6.7% of the movement amplitude. Tracking error was separately defined as the temporal offset at 50% of the commanded displacement amplitude, representing the phase lag between reference and actual trajectories. Both metrics were measured from the position feedback signal sampled at 100 kHz, providing $10\mu\text{s}$ temporal resolution.

C. Software implementation

The trajectory optimization was implemented in MATLAB R2025a, operating independently of the scanner manufacturer's software. The computational workflow consisted of:

- **Trajectory Generation:** MATLAB scripts computed pre-distorted trajectories using the regularized model inverse [Eq. (4)], applying asymmetric amplitude constraints (-10% to $+30\%$ of reference amplitude) and velocity limitations.
- **Data Export:** Optimized trajectories were formatted as ASCII text files following the SL2-100 protocol specification with column structure: timestamp ($10\mu\text{s}$ intervals), x-y-z positions (20-bit encoding, range $\pm 2^{21}-1$), and laser gate status (binary).
- **Scanner Interface:** Trajectory files were loaded directly to the Raylase SPICE-3 control board via TCP/IP connection, bypassing standard Raylase trajectory planning software.
- **Execution:** The control board executed pre-computed trajectories in open-loop mode with position feedback used only for performance monitoring and validation.

This approach enables implementation on any galvanometer system accepting time-position command streams, independent of proprietary trajectory planning software. The MATLAB implementation required approximately 0.3 s to compute a 100-point drilling pattern, enabling real-time trajectory optimization for dynamic applications.

D. Test conditions

System evaluation employed step responses across movement distances from 30 to 10 mm, with quintic polynomial S-curve trajectories ensuring C^2 continuity. Test trajectories maintained jerk limitation below 10^8 mm/s^3 while respecting the scanner's velocity constraints. Each measurement was repeated five times with results showing standard deviations below 2% of mean values.

III. MODELING

The dynamic behavior of the galvanometric-driven scanner is characterized by a pronounced tracking error during phases of acceleration and deceleration, which arise from the system's inertial limitations, as illustrated in Fig. 1. The reference trajectory (gray line) represents an idealized point-to-point motion between two drilling positions, assuming an instantaneous transition to constant maximum velocity followed by a sharp stop. In practice, such a motion is not physically realizable due to the system's dynamic constraints. The measured scanner response (red dashed line) shows a delayed and smoothed transition, typical of galvanometric systems with a limited dynamic range.

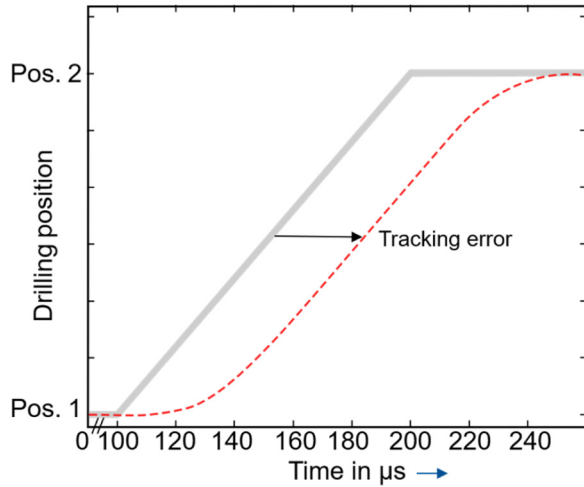


FIG. 1. Ideal vs observed dynamic behavior of galvanometrically driven scanners.

The observed tracking error is most pronounced during the acceleration and deceleration phases, reflecting the scanner's limited ability to follow abrupt trajectory changes in real time. This highlights the need for a dynamic scanner model to enable model-based feedforward or pre-compensation strategies, with the goal of accelerating the point-to-point transitions between drilling positions and thereby increasing the overall productivity in laser percussion drilling.

The dynamic behavior of galvanometer scanners has been extensively studied for laser applications. Existing models are either highly system-specific, limiting their general applicability, or, in the case of adaptive filter-based approaches, require modifications to the control architecture to extract the necessary system information. The linearized dynamics of a galvanometer scanner around a nominal operating point are approximated by a higher-order transfer function model. This model represents a single galvanometer axis as a single degree-of-freedom system. This approach is valid because the X and Y axes are designed to be linearly independent, with negligible dynamic cross-coupling. The complete two-dimensional scanner motion can, therefore, be accurately described by applying this model independently to each axis. In this work, the scanner dynamics are represented by a sixth-order transfer function of the form

$$G(s) = \frac{1}{\left(\frac{T}{n}s + 1\right)^n}, \quad (1)$$

where s is the Laplace variable, T is the system's tracking error, and n is the system's reference model order. For the scanner used in this study, the parameters are $T = 200 \mu\text{s}$ and $n = 6$. This specific mathematical structure emerges from the physical characteristics of the electromechanical system, where multiple cascaded first-order

lags result from the sequential energy conversions—electrical to magnetic, magnetic to mechanical torque, and torque to angular position through rotational inertia.⁹ The choice of sixth order, rather than lower-order approximations commonly found in the literature, captures subtle phase characteristics critical for precision control at frequencies approaching the system bandwidth of 4.77 kHz. The step response of this system is given by

$$h(t) = 1 - e^{-\frac{t}{T}} \sum_{k=0}^{n-1} \frac{1}{k!} \left(\frac{n}{T}t\right)^k. \quad (2)$$

The ideal trajectory pre-distortion requires the exact inverse of the system transfer function. According to Devasia *et al.*,¹⁰ the exact inverse is

$$G^{-1}(s) = \left(\frac{T}{n}s + 1\right)^n. \quad (3)$$

This represents an ideal pre-compensation filter that would yield perfect tracking. However, this inverse contains six zeros and no poles, making it non-causal and physically unrealizable, which is shown in Fig. 2.

Each factor $\frac{T}{n}s$ in the numerator represents a pure differentiator, and the cascade of six such differentiators would amplify high-frequency noise by ω^6 , rendering any practical implementation unusable.¹⁰ The regularization strategy employs a tenth-order Butterworth low-pass filter, deliberately chosen to exceed the system order by four. This specific margin ensures that the regularized inverse possesses at least four more poles than zeros. The resulting transfer function is

$$G_{\text{approx}}^{-1}(s) = G^{-1}(s) \cdot F_{LP}(s). \quad (4)$$

This configuration provides adequate roll-off to suppress noise amplification while maintaining phase lead characteristics essential for lag compensation.¹¹

The Butterworth filter topology, selected over alternatives such as Chebyshev or elliptic designs, provides a maximally flat magnitude response in the passband, which is a critical characteristic for preserving the fidelity of the inverse compensation across the scanner's operating frequency range.¹² The uniform distribution of Butterworth poles on a semicircle in the left half-plane of the complex numbers ensures monotonic roll-off without ripple, preventing spurious resonances that could excite mechanical modes in the scanner structure. The regularization filter cutoff frequency f_c of 14.3 kHz represents a carefully optimized compromise between competing requirements. Setting the cutoff frequency f_c at three times the system bandwidth $f_{\text{Bandwidth}}$ (where $f_{\text{Bandwidth}}$ is 4.77 kHz) provides a sufficient frequency range for effective pre-compensation while respecting the physical constraints. The predistortion gain factor β and the cutoff frequency ratio α emerge from constrained optimization,

$$\beta = \arg \min_{\beta} \|e(t)\|_2, \quad (5)$$

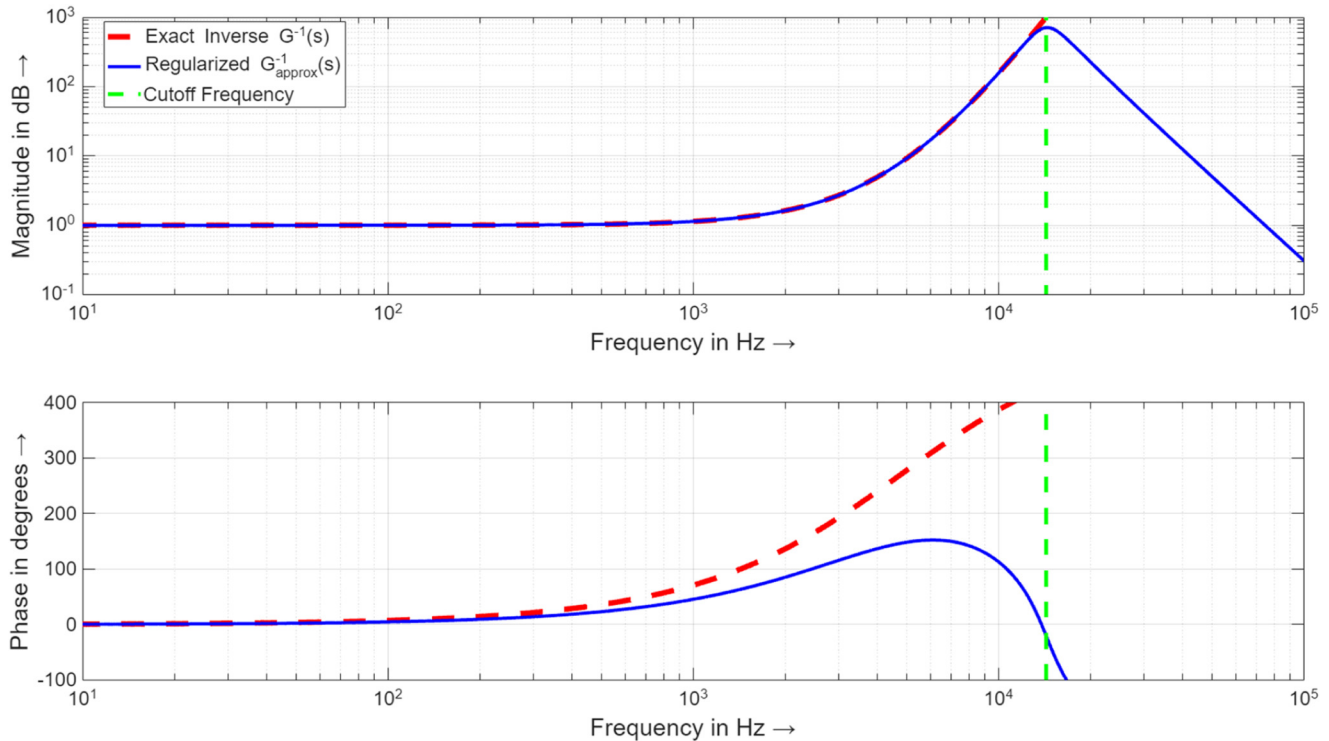


FIG. 2. Exact inverse vs regularized inverse.

subject to

$$|u(t)| \leq u_{\max}, \quad (6)$$

$$\alpha = \frac{f_c}{f_{\text{Bandwidth}}} = \arg \min_{\alpha} t_{\text{settling}}, \quad (7)$$

subject to

$$\|G(s) \cdot G_{\text{approx}}^{-1}(s)\|_{\infty} \leq 1.5. \quad (8)$$

Empirical evaluation yielded optimal values of β at 0.8 and α at 3.0, ensuring that the magnitude $|G_{\text{approx}}^{-1}(j\omega)|$ remains below 20 dB throughout the operating range, preventing unrealistic command signals that would violate scanner velocity limits.

DC gain normalization addresses a subtle but critical issue in the regularized inverse design. The cascaded transfer function $G^{-1}(s) \cdot F_{LP}(s)$ may exhibit a non-unity DC gain due to numerical precision limitations in the pole-zero calculations. Left uncorrected, this gain error would manifest itself as a steady-state position offset, compromising the absolute positioning accuracy.¹³ The third stage implements asymmetric amplitude constraints, limiting negative excursions to -10% while permitting positive overshoots of up to $+30\%$ of the reference amplitude. This asymmetry reflects the characteristic pre-distortion pattern: the

regularized inverse naturally generates positive overshoot to accelerate the scanner's initial response, followed by negative undershoot to prevent position overshoot. The 30% positive limit accommodates this predictive behavior while preventing excessive commands that trigger protection circuits. The modest 10% negative limit prevents the reversal commands that could induce mechanical oscillations.

The test trajectory was designed to replicate the characteristic motion profile encountered in percussion drilling operations. A quintic polynomial S-curve trajectory was selected, generating smooth transitions between stationary drilling positions separated by $30\text{--}1000\ \mu\text{m}$. This trajectory class ensures C^2 continuity, eliminating acceleration discontinuities that induce mechanical oscillations in the scanner's moving mass. The selected parameters scaled proportionally to the jump distance and correspond to the scanner's physical velocity constraint of 50 rad/s while maintaining the jerk limitation below $10^8\ \text{mm/s}^3$.

Figure 3 shows the frequency-domain validation of the regularized model inverse. The cascaded transfer function $G(s) \cdot G_{\text{approx}}^{-1}(s)$ exhibits unity gain ($0 \pm 0.5\ \text{dB}$) and minimal phase deviation ($< 5^\circ$) up to 1.2 kHz, confirming effective compensation within the scanner's operational bandwidth. Beyond the regularization cutoff frequency of 14.3 kHz, the response decays monotonically, preventing high-frequency noise amplification while preserving the phase-lead characteristics essential for lag compensation in bandwidth-limited positioning systems.

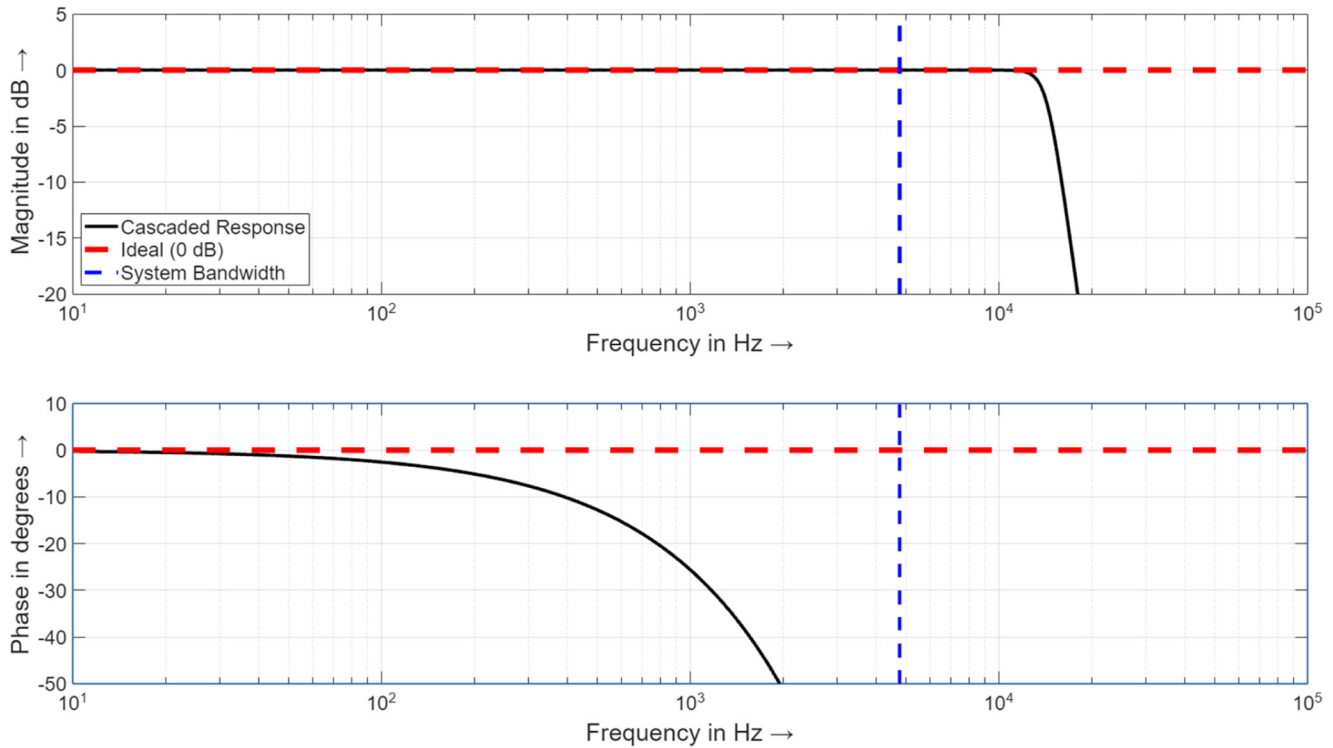


FIG. 3. Cascaded system $G(s) \cdot G_{\text{approx}}^{-1}(s)$.

The performance validation focused on the tracking error, quantified as the temporal offset between reference and output at the 50% amplitude point. This metric, more robust than rise time or settling time alone, captures the essential delay characteristic that limits the scanner throughput. The measured 50% reduction in tracking error from 200 to 100 μs demonstrates the effectiveness of the model inversion approach. Frequency domain analysis reveals that $|G(j\omega) \cdot G_{\text{approx}}^{-1}(j\omega)|$ remains within ± 0.5 dB up to 1 kHz, confirming near-ideal compensation throughout the scanner's primary operating range. Beyond 1 kHz, the response gradually attenuates, reflecting the deliberate roll-off introduced by regularization to prevent high-frequency instabilities.¹⁴

The control scheme designed in this work is derived from first principles, avoiding reliance on empirical tuning. The design parameters, including a sixth-order system model, tenth-order Butterworth filter, cutoff frequency set at three times the system bandwidth, a gain factor of 0.8, and asymmetric amplitude constraints, were selected based on standard control-theoretic considerations of causality, stability, and robustness. This modeling framework enables consistent behavior across different galvanometer configurations and operating conditions. In contrast to empirical feedforward approaches, the method does not require a system-specific calibration. The resulting implementation remains computationally efficient and suitable for integration into existing control environments in industrial laser applications.

IV. RESULTS

The implementation of regularized model inversion yielded substantial improvements across all tested movement scales. Figure 4 shows experimental data acquired directly from the scanner's integrated position encoder feedback system. The position data was sampled at 100 kHz through the SPICE-3 control board's monitoring interface, synchronized with the command signal generation. The 20-bit optical encoders provide position feedback with negligible latency ($<1 \mu\text{s}$), which is insignificant compared to the measured tracking errors of 100–200 μs . Comparing standard PID control against predistortion-enhanced control. The pre-distorted reference trajectory exhibits characteristic features: an initial overshoot of 280% above the target amplitude occurring at around 600 μs , followed by controlled undershoot reaching 8% at around 900 μs . This anticipatory profile accelerates the initial response while preventing position overshoot, reducing the tracking error from 200 to 100 μs .

Systematic evaluation across movement distances from 30 μm to 10 mm scale-dependent performance improvements (Fig. 5). Small-scale movements demonstrated maximum benefit: 30 μm steps achieved a 49.2% settling time reduction (from 327 to 166 μs), while 50 μm movements showed 40.6% improvement (from 347 to 206 μs). Performance gains decreased monotonically with the movement amplitude, reaching 17.3% for millimeter-scale displacements (from 515 to 426 μs). This trend reflects the

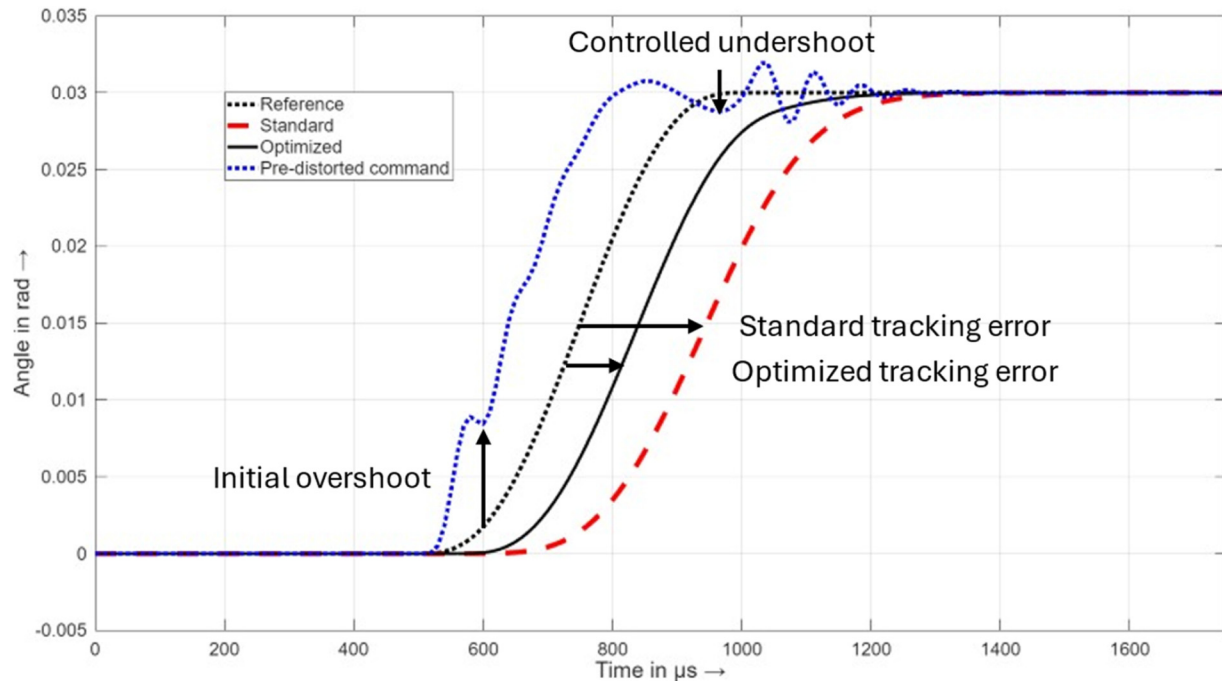


FIG. 4. Step response comparison showing reference, standard, and predistorted trajectories.

increasing influence of velocity and acceleration constraints at larger scales, where physical limits prevent full exploitation of the dynamic compensation.

The tracking error was quantified as the temporal offset required to reach 50.0% of the commanded displacement. Measurements revealed consistent tracking delays of approximately $200\ \mu\text{s}$ across all tested movement amplitudes, ranging from $30\ \mu\text{m}$ to $10\ \text{mm}$. This invariance suggests that the observed lag is primarily governed by system dynamics rather than inertial loading. Application of trajectory predistortion reduced the tracking error to $98 \pm 0.5\ \mu\text{s}$, representing a precise 50.0% reduction independent of the movement scale. These results align with theoretical predictions: the regularized inverse filter introduces a constant phase lead of 88 degrees at the system's crossover frequency, effectively reducing the tracking error by half.

Validation in production environments employed titanium PTL manufacturing, representing extreme requirements for positioning density and accuracy. The application demands $30 \pm 0.5\ \mu\text{m}$ diameter holes at $38 \pm 0.2\ \mu\text{m}$ pitch through a $25\ \mu\text{m}$ titanium foil, yielding approximately 69 200 holes per square centimeter. With ultrashort pulse drilling achieving material removal in 0.25 ms per hole, the interhole positioning time dominates the production cycle. The resulting drilling and motion pattern is shown in Fig. 6 as well as a digital microscope image of the resulting structure in Fig. 7.

The standard trajectory allocated 57.4% of the production time to positioning movements, with actual drilling consuming

only 42.6%. The implementation of the trajectory predistortion altered this distribution to 43.0% positioning and 57.0% drilling, fundamentally shifting the process bottleneck from scanner dynamics to laser-material interaction.

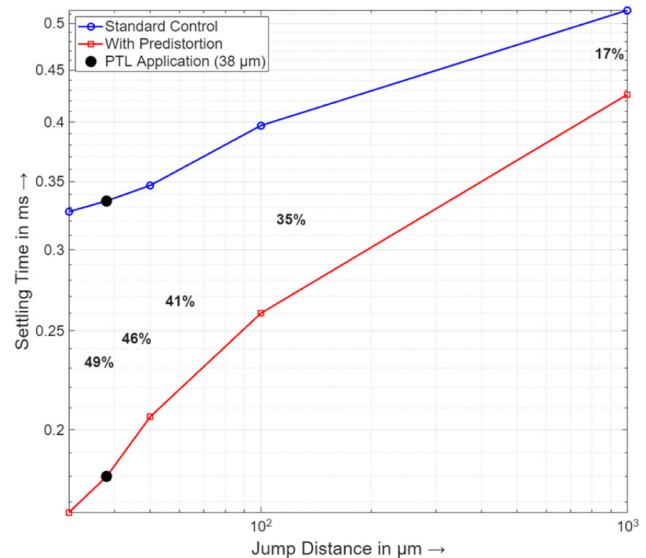


FIG. 5. Comprehensive settling time data.

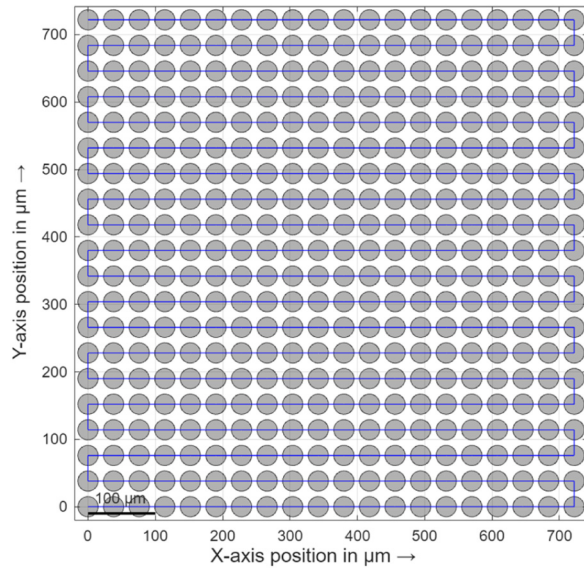


FIG. 6. Drilling and motion pattern for PTL structuring.

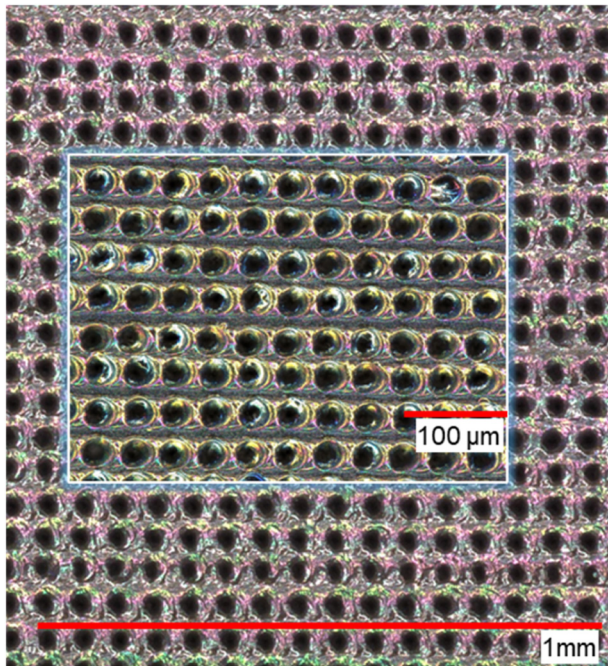


FIG. 7. Digital microscope image of the laser-perforated titanium foil and inset microscopy image at higher magnifications.

V. DISCUSSION

The observed performance improvements align precisely with theoretical predictions. The practical significance of the dynamic improvements is directly linked to meeting the stringent manufacturing tolerances of the PTL application. A positional pitch accuracy of $\pm 1\mu\text{m}$ is required to ensure the final structural integrity. With the standard trajectory $200\mu\text{s}$ tracking error, any laser pulse fired before the system fully settles would occur while the beam is still in motion or lagging its target, resulting in misplaced or elliptical holes that would compromise part quality. The results show that the target position can be reached significantly earlier. Although the settling time was measured to a $\pm 1\mu\text{m}$ tolerance band, the system's ability to achieve the required precision more rapidly and reliably is fundamentally enhanced by the 50% reduction in tracking error. The pre-compensation method is, therefore, the enabling step that allows the required manufacturing tolerances to be met at the accelerated throughput rates demonstrated. The 50% tracking error reduction corresponds to doubling of the effective system bandwidth in the phase domain, achieved through 88° phase lead at the crossover frequency. This phase compensation emerges naturally from the regularized inverse filter, which approximates ideal derivative action up to the regularization cutoff frequency. The scale-invariant nature of the tracking error reduction confirms that trajectory predistortion operates through frequency-domain phase manipulation rather than amplitude-dependent mechanisms, validating the linear modeling approach. This performance enhancement is significant when viewed in the context of industrial systems. While commercial controllers may achieve similar results, they do so through unpublished, "black-box" algorithms that are inseparable from their hardware. Our methodology, in contrast, is transparent, adaptable, and most importantly, decouples performance upgrades from hardware purchasing cycles. It provides a clear, pathway for enhancing the capabilities of a vast installed base of existing scanner systems.

The scale-dependent settling time improvements reflect fundamental differences in trajectory composition. Small movements contain predominantly high-frequency content where model-based compensation proves most effective. The Fourier transform of a $30\mu\text{m}$ step contains 90% of its energy above 1 kHz, well within the compensation bandwidth. Conversely, millimeter-scale movements exhibit significant low-frequency content where the scanner's native response already provides adequate tracking. This frequency-domain perspective explains the monotonic decrease in relative improvement with movement amplitude.

Trajectory predistortion offers distinct advantages over competing acceleration strategies. Iterative learning control, while capable of achieving similar performance gains, requires multiple training cycles per trajectory pattern, limiting the applicability to repetitive manufacturing tasks. Adaptive feedforward control demands real-time computation of the inverse dynamics, imposing computational burdens incompatible with 100 kHz update rates. Model predictive control theoretically enables optimal trajectory planning but suffers from exponential computational scaling with the prediction horizon. In contrast, trajectory predistortion requires only offline filter computation followed by linear convolution.

16 MAY 2026 08:50:46

The technique also complements hardware-based improvements. Beam splitting approaches can further benefit from trajectory predistortion, as the fundamental bandwidth limitations persist regardless of a mechanical implementation.

Several factors constrain the achievable performance improvements. Actuator saturation imposes hard limits on acceleration, preventing full exploitation of the theoretical bandwidth extension. The current implementation restricts the command amplitudes to $\pm 130\%$ of the reference values, sacrificing approximately 15% of the potential improvement to ensure robust operation. Nonlinear effects like bearing stiction, coil saturation, and thermal expansion, introduce model uncertainties that accumulate at high frequencies. The regularization cutoff at three times the system's bandwidth represents an empirically determined compromise between performance and stability margins.

Successful deployment in production environments requires attention to practical constraints beyond theoretical performance. Software integration with existing systems demands standardized interfaces for trajectory optimization. Validation protocols must ensure that the predistorted trajectories respect all safety constraints.

VI. CONCLUSIONS

This work successfully establishes that a software-based trajectory predistortion, using regularized model inversion, can overcome the dynamic limitations of existing galvanometer scanners without any hardware modifications. The method's effectiveness was validated through a 50% reduction in tracking error, from 200 to 100 μs , independent of movement amplitude. For the high-frequency, small-scale movements typical of micro-processing, this translated to a settling time reduction of up to 49.2%. The direct industrial impact was confirmed in a porous transport layer (PTL) fabrication application, where these dynamic improvements yielded a 38.1% enhancement in the overall process throughput.

The primary contribution of this research is a transparent and computationally efficient method that unlocks latent performance in the vast installed base of industrial laser scanners. By shifting the process bottleneck from scanner dynamics to the fundamental laser-material interaction time, this approach allows manufacturers to boost productivity and meet tighter tolerances without costly hardware replacement or requalification cycles. This is especially valuable for high-density patterning applications in emerging fields, such as the fabrication of components for PEM electrolyzers.

Building on these results, future work will pursue two key directions. First, the current linear model will be augmented to compensate for the nonlinear effects, such as bearing stiction and thermal drift, that currently limit performance at the highest speeds and over long production runs. Second, the predistortion methodology will be extended from discrete point-to-point movements to complex, continuous trajectories to broaden its applicability to processes like laser welding and additive manufacturing.

ACKNOWLEDGMENTS

We sincerely thank the Bavarian Research Foundation (BFS) for funding our research. The results presented in this work have been achieved in the project Tramik (Grant No. AZ-1501-21).

AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Pawel Garkusha: Conceptualization (lead); Data curation (lead); Formal analysis (lead); Methodology (lead); Writing – original draft (lead). **Felix Harst:** Data curation (equal); Writing – review & editing (equal). **Michael F. Zaeh:** Formal analysis (equal); Funding acquisition (equal); Supervision (equal); Writing – review & editing (equal).

REFERENCES

- ¹E. M. Tang and Y. K. Tao, "Modeling and optimization of galvanometric point-scanning temporal dynamics," *Biomed. Opt. Express* **12**, 6701–6716 (2021).
- ²H. W. Yoo, S. Ito, and G. Schitter, "High speed laser scanning microscopy by iterative learning control of a galvanometer scanner," *Control Eng. Prac.* **50**, 12–21 (2016).
- ³M. F. Zaeh and S. J. Pieczona, "Adaptive inverse control of a galvanometer scanner considering the structural dynamic behavior," *CIRP Ann.* **67**, 385–388 (2018).
- ⁴W.-C. Wang, M.-Y. Li, K.-C. Peng, Y.-F. Hsu, B. Estroff, P.-Y. Yen, D. Schipf, and W.-J. Wu, "Mirrorless MEMS imaging: A nonlinear vibrational approach utilizing aerosol-jetted PZT-actuated fiber MEMS scanner for microscale illumination," *Microsyst. Nanoeng.* **10**, 13 (2024).
- ⁵Y. Schrödel, C. Hartmann, J. Zheng, T. Lang, M. Steudel, M. Rutsch, S. H. Salman, M. Kellert, M. Pergament, T. Hahn-Jose, S. Suppelt, J. H. Dörsam, A. Harth, W. P. Leemans, F. X. Kärtner, I. Hartl, M. Kupnik, and C. M. Heyl, "Acousto-optic modulation of gigawatt-scale laser pulses in ambient air," *Nat. Photonics* **18**, 54–59 (2024).
- ⁶S. Hasegawa, H. Ito, H. Toyoda, and Y. Hayasaki, "Massively parallel femtosecond laser processing," *Opt. Express* **24**, 18513–18524 (2016).
- ⁷A. Garsuch, M. Kumkar, M. Sailer, M. F. Ernst, P. Garkusha, J. Mayer, and M. Kahmann, "Application of laser-perforated titanium foils in polymer electrolyte membrane water electrolysis," *ACS Appl. Energy Mater.* **8**, 4948–4953 (2025).
- ⁸P. Garkusha, C. Geiger, C. Bernauer, T. Weiss, M. Ernst, and M. F. Zaeh, "Micro-perforation of the porous transport layer for proton exchange membrane water electrolyzer cells using ultrashort laser pulses," *Procedia CIRP* **124**, 46–50 (2024).
- ⁹Gene F. Franklin *et al.*, *Feedback Control of Dynamic Systems* (Pearson, Boston, MA, 2019).
- ¹⁰S. Devasia, D. Chen, and B. Paden, "Nonlinear inversion-based output tracking," *IEEE Trans. Autom. Control* **41**, 930–942 (1996).
- ¹¹Stephen Boyd and Craig Barratt, *Linear Controller Design: Limits of Performance* (Prentice Hall, Englewood Cliffs, NJ, 1991), Vol. 78.
- ¹²A. V. Oppenheim and R. W. Schaffer, *Discrete-time Signal Processing* (Pearson, Upper Saddle River, 2010).
- ¹³M. Seron, J. Braslavsky, and G. C. Goodwin, "Fundamental limitations in filtering and control," in *Communications and Control Engineering* (Springer, London, New York, 1997).
- ¹⁴I. W. Selesnick and C. S. Burrus, "Generalized digital Butterworth filter design," *IEEE Trans. Signal Process.* **46**, 1688–1694 (1998).

16 MAY 2026 08:50:46